

ECE 315: Signals and Systems

Homework 4

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February 20, 2026

This homework is due on **2/27, 11:59pm**. For analytical problems, write your solutions independently by hand. For coding problems, attach your source code and answers. Combine them in one PDF document and submit on CANVAS. Please ensure that your PDF is clearly readable on a computer.

1. (2pts) Consider the following discrete-time systems with input $x[n]$ and output $y[n]$:
 - $y[n] = x[n - 1] + nx[n]$
 - $y[n] = (x[n] + x[n - 1])^2$
 - $y[n] = \max(x[n], x[n - 1], x[n - 2])$
 - $y[n] = \min(x[n], -x[n - 1])$
 - (i) Determine whether they are linear and time-invariant, and compute their impulse and step responses (i.e., $x[n] = \delta[n]$ and $u[n]$).
 - (ii) If we pretend each system is linear and time-invariant, its impulse response completely captures the system behavior. With this assumption, compute the system response for each case when the input is $x[n] = u[n]$ and answer if it is different from the original step response for each case.
2. (2pts) Consider the following continuous-time systems with input $x(t)$ and output $y(t)$:
 - $y(t) = x(t - 1) + tx(t)$
 - $y(t) = (x(t) + x(t - 1))^2$
 - $y(t) = \max(x(t), x(t - 1), x(t - 2))$
 - $y(t) = \min(x(t), -x(t - 1))$
 - (i) Determine whether they are linear and time-invariant, and compute their impulse and step responses (i.e., $x(t) = \delta(t)$ and $u(t)$).

- (ii) If we pretend each system is linear and time-invariant, its impulse response completely captures the system behavior. With this assumption, compute the system response for each case when the input is $x(t) = u(t)$ and answer if it is different from the original step response for each case.
3. (2pt) Given a continuous-time linear and time-invariant system's impulse response $h(t)$, we can compute the system's output for an α_i -weighted sum of m input signals: $x(t) = \sum_{i=1}^m \alpha_i x_i(t)$ through a α_i -weighted sum of convolution integrals. Prove that

$$h(t) * x(t) = \sum_{i=1}^m \alpha_i h(t) * x_i(t). \quad (1)$$

We consider the following impulse response and three inputs ($m = 3$):

- $h(t) = \begin{cases} 2^{-t} & \text{when } 0 \leq t \leq 4 \\ -2^t & \text{when } -2 \leq t < 0 \end{cases}$
- $x_1(t) = u(t)$, $x_2(t) = u(t+2)$, and $x_3(t) = u(-t)$.

Compute the system's output for the input $x[n] = \sum_{i=1}^3 \alpha_i x_i(t)$ using (1) when $\alpha_1 = 1$, $\alpha_2 = \frac{1}{2}$, and $\alpha_3 = 2$.

4. (2pt) Consider a discrete-time linear time-invariant system with impulse response $h_1[n] * h_2[n]$ and input $x[n]$:

$$y[n] = x[n] * (h_1[n] * h_2[n]).$$

- (i) Prove that the associative property: $x[n] * (h_1[n] * h_2[n]) = (x[n] * h_1[n]) * h_2[n]$, and compute the output when $h_1[n] = \sin(2n)$, $h_2[n] = 2^{-n}u[n]$, and $x[n] = \delta[n] - \delta[n-2]$.
- (ii) Come up with an example that shows a non-LTI system does not have the associative property.
5. (2pts) Consider the following impulse responses of discrete- and continuous-time linear time-invariant systems:

- $h[n] = 2^n u[4-n]$
- $h[n] = n(\frac{1}{2})^n u[n-1]$
- $h[n] = \frac{1}{2^n} u[4-n]$
- $h[n] = (-\frac{1}{2})^n u[n] + (\frac{3}{2})^n u[1-n]$
- $h(t) = e^{-2t} u(t-1)$
- $h(t) = e^{-4t} u(2-t)$
- $h(t) = te^{-t} u(t)$
- $h(t) = 2e^{-t} u(t) + e^t u(1-t)$

Determine the causality and stability for each system described above.

6. Read Section **2.7**, and complete the MATLAB problems **2.7-2** and **2.7-5** in **LSS**.